K-P-146D
September 6, 1968
SUPERSEDING
Fed. Spec. K-P-146C
May 6, 1963

FEDERAL SPECIFICATION

TARPAULINS: COTTON DUCK; FWWMR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers uncoated cotton duck tarpaulins (see 6.1).

1.2 Classification.

1.2.1 Types. Tarpaulins shall be of the following types, as specified (see 6.3):

Type I - With loops.
Type II - With grommets and tie ropes.

1.2.2 Classes and sizes. Tarpaulins shall be of the following classes and sizes as specified (see 6.3):

Class 1 - Cloth, duck, cotton, No. 6 hard texture, olive drab.

Size
5 feet by 5 feet.
10 feet by 6 feet.
12 feet by 10 feet.
17 feet by 12 feet.
20 feet by 20 feet.
23 feet by 15 feet.
26 feet by 22 feet.
40 feet by 20 feet.

Class 2 - Cloth, duck, cotton, No. 6 hard texture, deck gray Navy shade 3104.

Size
8 feet by 8 feet.
15 feet by 15 feet.
20 feet by 20 feet.

Class 3 - Cloth, duck, cotton, Nos. 10 or 11 hard texture, unbleached.

Size
7 feet by 15 feet.
7 feet by 16 feet.
7 feet by 18 feet.
16 feet by 16 feet.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

T-R-605 - Rope, Manila and Sisal.

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T-T-616 - Treatment: Mildew Resistant for Rope and Cord.
T-T-871 - Twine, Cotton, Wrapping.
T-T-911 - Twine, Jute.
V-T-285 - Thread, Polyester.
CCC-C-419 - Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
CCC-C-428 - Cloth, Duck, Cotton; Fire, Water, Weather, and Mildew Resistant.
CCC-C-429 - Cloth, Osnaburg, Cotton.
CCC-C-467 - Cloth, Burlap, Jute (or Kenaf).
DDD-L-20 - Label; for Clothing, Equipage, and Tentage, (General Use).
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Box, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Box, Wood, Nailed and Lock-Corner.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 751 - Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Washington, D. C., Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.)

(Federal Government activities may obtain copies of Federal specifications, standards, and handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-W-530 - Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-M-13295 - Mildew Resistant Compound, Textile (for Field Retreatment).
MIL-G-16491 - Grommet, Metallic.
MIL-T-40625 - Tubing, Bias Sewn (Burlap or Osnaburg) Cloth.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Drawing:

Army Natick Laboratories

5-10-9 - Tarpaulins: Duck, Cotton; FWWMR.

(Figure 1 is a minature reproduction of referenced drawing and is attached for information only.)

(Copies of Military specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Department of the Army Technical Manual TM10-269 - General Repair for Canvas and Webbing.

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National Motor Freight Classification

(Application for copies should be addressed to the Consolidated Classification Board, 1616 P Street, N. W., Washington, D. C. 20036.)

Uniform Freight Classification

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.3), before production is commenced, a sample tarpaulin covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production item.

3.2 Materials and components.

3.2.1 Cloth, duck, cotton, olive drab. The cloth for class 1 tarpaulins shall be type I, No. 6, hard texture of CCC-C-419, finished to conform to type I, class 1 of CCC-C-428.

3.2.2 Cloth, duck, cotton, deck gray Navy shade 3104. The cloth for class 2 tarpaulins shall be type I, No. 6, hard texture of CCC-C-419, finished to conform to type II, class 1 of CCC-C-428.

3.2.3 Cloth, duck, cotton, unbleached. The cloth for class 3 tarpaulins shall be type I, Nos. 10 or 11, hard textured duck of CCC-C-419, except for weight which shall be 12.75 ounces per square yard minimum and 14.35 ounces per square yard maximum. Air permeability requirements shall not be applicable. The unbleached cloth shall be finished to conform to class 1 of CCC-C-428, except that the flexibility requirements shall not apply.

3.2.4 Webbing, textile, cotton. Webbing shall conform to type III, class 4, 1-inch width of MIL-W-530, dyed shade olive drab No. 7 for class 1 tarpaulins and dyed deck gray Navy shade 3104 for class 2 tarpaulins. When type I, class 3 tarpaulins are required, webbing shall conform to type III, class 1b of MIL-W-530.

3.2.5 Thread, polyester. Thread shall conform to type I, class 1, subclass B of V-T-285. Sizes shall be as follows:

For all stitching except zig-zag stitching and stitching indicated by "Z" on drawing.	-	Size FF for needle and bobbin.
For zig-zag stitching.	-	Size F for needle and bobbin.
For stitching indicated by "Z" on drawing.	-	Size FF for needle and size F for looper.

3.2.5.1 Color and treatment. Thread for the class 2 and 3 tarpaulins shall be natural color. Thread for the class 1 tarpaulins shall be dyed shade S-1 (Cable No. 66022). Dyed thread shall show colorfastness to weathering equal to or better than the standard shade sample (see 6.4). If no standard equal to or better than the standard shade sample is available, the thread shall show good colorfastness to weathering.

3.2.6 Twine, cotton. Twine shall conform to type II, class 1, 5-ply of T-T-871.

3.2.7 Rope, manila. Rope shall conform to type M, class 2 of T-R-605, 1-1/8 inch circumference, mildew-resistant treated in accordance with T-T-616.

3.2.8 Grommet, metallic. Grommets shall conform to type III, class 3, size No. 5 of MIL-G-16491.

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3.3 Construction. Construction shall conform to Drawing 5-10-9 and as specified herein.

3.3.1 Stitching, machine.

3.3.1.1 Types. Stitch types shall conform to Fed. Std. No. 751 as follows:

For all stitching except zig-zag stitching and stitching indicated by "Z" on drawing.	-	Type 301, 5 to 7 stitches per inch.
For zig-zag stitching.	-	Type 304, 12 to 16 stitches per inch.
For stitching indicated by "Z" on drawing.	-	Type 401, 5 to 7 stitches per inch.

3.3.1.2 Thread breaks. Thread breaks in stitching shall be overstitched not less than 1-inch on stitch type 301, not less than 3/4-inch on stitch type 304 and not less than 1-1/2 inches at each break in stitch type 401. Thread breaks in type 401 stitching may be overstitched with stitch type 301.

3.3.1.3 Stitching ends. The ends of type 301 stitching shall be overstitched not less than 1-inch except where ends are turned under in a hem or held down by other stitching. Where stitching is performed automatically on stitch patterns such as box, box with cross-stitch, "W" stitching or straight line tacking, 3 or more tying, overlapping or backstitches shall be used to secure the ends.

3.3.1.4 Skipped stitches. Two or more consecutively skipped stitches occuring in types 301 or 304 stitching shall be overstitched not less than 1-inch. Any skipped stitches in type 401 stitching shall be overstitched not less than 1-1/2 inches. Skipped stitches in 401 stitching may be overstitched with type 301 stitching.

3.3.1.5 The addition of any lubricant to the polyester thread prior to or during the sewing operation is prohibited.

3.3.2 Setting of grommets, type II tarpaulins. Holes punched to receive the grommets shall be smaller than the grommet barrel so that the barrel must be forced through the hole. The grommets shall be securely clinched without cutting the materials.

3.3.3 Tie ropes, type II tarpaulins. The tie ropes shall be constructed of the material specified in 3.2.7. Eye splices shall have 3 tucks and the ends of all strands shall be clipped to a length of not less than one inch nor more than 1-1/4 inches. The size of the eye shall be as specified on the drawing and shall be the measurement of the inside of the loop when stretched taut. The free ends of the tie ropes shall be either machine or hand served as shown on the drawing. Hand serving shall be done with twine specified in 3.2.6, using two strands twisted together. Waxed twine shall be allowed.

3.3.4 Compliance with cup test. There shall be no seepage of water through the panel lap seam stitching when the tarpaulin is tested as specified in 4.4.1. The cup test is intended to insure that nonwicking thread has been utilized and that no lubrication has been added to the polyester thread during the sewing operation.

3.3.5 Repairs. Any necessary repairs shall be effected in accordance with Department of the Army Technical Manual TM10-269, except that handsewn patches or cemented patches will not be acceptable. Repairs required as a result of failure of the cup test shall be performed by treating with a compound conforming to type I of MIL-M-13295 prior to retesting (see 6.6).

3.4 Marking. Identification marking shall be applied in the location shown on the drawing and shall conform to type IV, class 8 of DDD-L-20. The letters "US" shall be applied in the size characters and in the location shown on the drawing and shall conform to type IV, class 9 of DDD-L-20. Class 9 markings shall be clearly legible after subjection to accelerated weathering.

3.5 Workmanship. Cloth components shall be clean and free of holes, cuts, tears, or cloth defects such as multiple floats or broken or missing yarns. Webbing shall contain no frayed or scalloped edges and rope components shall be free of cuts or breaks. Thread tension shall be maintained so that there will be no loose stitching and seam allowances shall be maintained

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with seams properly sewed so that no runoffs, twists, pleats, or open seams shall result. Care shall be taken in sewing to see that no needle chews occur. All thread ends shall be trimmed to 1/2 inch or less. Metal components shall be free of burrs, sharp edges or corroded areas and shall not be broken or malformed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. When a preproduction sample is required, it shall be examined for defects listed in table I and dimensions specified.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 Process inspection. Inspection shall be made to determine that the following are as specified:

a. There shall be no lubrication of the polyester thread prior to or during the sewing operations for compliance with 3.3.1.5.

b. Holes punched to receive grommets are in compliance with 3.3.2.

Whenever nonconformance is noted, correction shall be made to the items affected and the process.

4.3.3 End product inspection. The lot shall consist of all completely fabricated tarpaulins offered for inspection at one time. The sample unit shall be one completely fabricated tarpaulin. The inspection levels and acceptable quality levels (AQLs) shall be as specified in 4.3.3.3.

4.3.3.1 Visual examination. Visual examination of the completely fabricated tarpaulin shall be made for defects outlined in table I.

TABLE I. Classification of defects for visual characteristics

Examine	Defect	Classification		
		Major	Minor	
			A	B
Cloth	Cut, tear or hole (including exposed drill hole):			
	- greater than 1/8 inch in length or diameter.	X		
	- 1/8 inch or less in length or diameter:			
	- more than 10 holes.	X		
	- more than 5 but not more than 10 holes.		X	
	- 5 holes or less.			X
	Broken or missing yarn, multiple floats clearly visible at normal inspection distance (approximately 3 feet).		X	
Webbing	Edge frayed or scalloped.			X

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TABLE I. Classification of defects for visual characteristics (con.)

Examine	Defect	Classification		
		Major	Minor	
			A	B
Rope components	Any cut; any break.		X	
	Eye splice not as specified.			X
	Free end not finished as specified.			X
Seams and stitching	Open seam on stitch type 301:			
	- for more than 1/4 inch but not more than 1 inch.			X
	- for more than 1 inch.		X	
	Open seam on stitch type 401:			
	- one row of stitching open for 1 inch or less.			X
	- one row of stitching open for more than 1 inch.		X	
	- both rows of stitching open (not necessarily at the same location).	X		
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken, or when one or more skipped or runoff stitches occur. On double-stitched seams, a seam shall be classified as open when either one or both sides of a seam are open.			
	Thread breaks:			
	- overstitched less than 1-1/2 inches on stitch type 401.			X
	- overstitched less than 1 inch on stitch type 301.			X
	- overstitched less than 3/4 inch on stitch type 304.			X
	NOTE: Thread breaks not overstitched shall be classified as open seams.			
	Skipped stitches:			
	- overstitched less than 1-1/2 inches on stitch type 401.			X
	- overstitched less than 1 inch on stitch types 301 or 304 (when 2 or more skipped stitches occur consecutively).			X
	NOTE: Skipped stitches not overstitched shall be classified as open seams.			
	Needle chews resulting in cut, tear or hole (see Cut, tear or hole).			
	One or more required rows of stitching omitted (except on boxstitching).	X		
Boxstitching	Incomplete:			
	- one row of stitching omitted.		X	
	- two or more rows of stitching omitted.	X		

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TABLE I. Classification of defects for visual characteristics (con.)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)	Any boxstitching less than specified length by more than 1/8 inch.			X
Stitching ends	Overstitched less than one inch on stitch type 301 (except where ends are held down by other stitching, turned under in a hem or where stitching is performed automatically).		X	
	Secured with less than 3 tying, overlapping, or backstitches when automatic stitching is performed.		X	
Seams	Seam pleated.		X	
Seam type	Wrong seam type.	X		
Stitch type	Wrong stitch type.	X		
Stitch tension	Loose, resulting in an exposed bobbin or top thread for more than 6 inches.		X	
	Tight, as evidenced by puckering on seams for more than 6 inches.		X	
Stitches per inch	Less than minimum specified:			
	- one stitch.			X
	- two or more stitches.		X	
	More than maximum specified:			
	- one stitch.			X
	- two or more stitches		X	
	NOTE: Variation in the number of stitches per inch, caused by the operator's speeding up the machine and pulling the cloth in order to sew over heavy places or heavy seams or in turning corners shall be classified as follows: a. Within the minor "B" defect classification - no defect. b. Within the minor "A" defect classification - minor "B" defect.			
Stitching gage	Not as specified.			X
Stitching margin	Larger than specified by more than 3/16 inch:			
	- for more than 2 inches but not more than 4 inches in length.			X
	- for more than 4 inches in length.		X	
Components and assembly	Any component part omitted (unless otherwise classified herein).	X		

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TABLE I. Classification of defects for visual characteristics (con.)

Examine	Defect	Classification		
		Major	Minor	
			A	B
Components and assembly (cont'd)				
Hems	Twisted or pleated.			X
	Constructed with partial turn under when a double turn under is required.		X	
Panel	Adjacent panels spliced; number of splices more than specified; spliced pieces not joined as shown on drawing.		X	
Mends, darns or patches	Not authorized by the contracting officer or not in accordance with general procedures as outlined in Department of the Army Technical Manual TM10-269.	X		
Grommets	Broken	X		
	Malformed:			
	- fails to perform intended function.	X		
	- but will perform intended function.		X	
	Corroded area.			X
	Burr or sharp edge which may cause injury in handling or damage to cloth.	X		
	Improperly set; two or more teeth exposed.			X
	Clinched loosely allowing grommet to rotate in hole.			X
	Clinched excessively tight, cutting cloth.		X	
Marking: "US" and identification	Omitted, incorrect, illegible or misplaced.			X
Cleanness	Grease or oil stains			X
	Thread ends not trimmed to 1/2 inch or less.			X

4.3.3.2 Dimensional examination. Dimensional examination of the completely fabricated tarpaulin shall be made and any dimension not within the specified tolerance shall be classified as a defect.

4.3.3.3 Inspection levels and AQLs. Inspection levels and AQLs, expressed in terms of defects per hundred units, shall be as follows:

Examination	Inspection level	AQLs
For defects applicable to table I	II	6.5 Major. 15.0 Major and minor A defects combined. 40.0 Total; major, minor A and B defects combined.
For defects applicable to 4.3.3.2	S-3	6.5 One class.

4.3.3.4 Testing of the end product. Tarpaulin samples shall be tested as specified in 4.4.1

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for compliance with 3.3.4. The sample unit shall be one completely fabricated tarpaulin. The lot shall consist of completely fabricated tarpaulins offered for inspection and the inspection level shall be S-3. Failure shall be cause for rejection of the lot.

4.3.4 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking requirements are in compliance with section 5 of this specification. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of shipping containers offered for delivery at one time. The inspection level shall be S-2 and the AQL shall be 2.5 expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examine	Defect
Markings	Omitted, incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of cloth wrapping and loose stitching.
Contents	Not as specified.

4.3.5 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-585, PPP-B-601 or PPP-B-621, examination for closure and strapping shall be in accordance with the applicable specification.

4.4 Test.

4.4.1 Seam leakage test. The tarpaulin shall be tested in one location on a panel lap seam.

4.4.1.1 Procedure. Suspend the section of the tarpaulin containing the area to be tested between two supports (boxes, wooden horses) so as to allow both sides of the cloth to be viewed. The area to be tested shall then be cupped into a hoop 6 to 7 inches in diameter; e. g., an embroidery loop, with the seam running through the center of the cupped cloth. Pour into the cupped area approximately one pint of cold tap water. Allow to stand for 5 minutes. At the end of the period, examine the test area for seepage of water through the thread and along the entire portion of the stitching exposed to the cup test.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.3).

5.1.1 Level A. Each tarpaulin shall be laid out flat. Tie ropes, where applicable, shall be placed toward the center of the tarpaulin. Each tarpaulin shall then be folded neatly and compactly and securely tied with twine conforming to type I or II, 16 ply of T-T-871. The portion of the tarpaulin containing identification markings, if applicable, shall be on the outside of the folded tarpaulin. Care shall be taken to expel all excess air during the folding operations.

5.1.2 Level C. Tarpaulins shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.3).

5.2.1 Level A

5.2.1.1 Cloth-covered bundles. Tarpaulins of one type, class and size only, packaged as specified in 5.1, shall be packed in quantities as specified in table III, wrapped and sewn in cotton osnaburg cloth conforming to class 2 of CCC-C-429; or burlap cloth conforming to

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class 3 of CCC-C-467. The wrapping shall be securely hand sewn with type I or II, 16-ply cotton twine conforming to T-T-871, or type I, size No. 2 jute twine conforming to T-T-911, with approximately one stitch to the inch and every third stitch overhand knotted. Alternatively, the bottom and side may be machine sewn. A minimum of five inches of wrapping material shall be gathered together on each of the four corners and securely hand sewn into ears for handles. In lieu of the sewn wrappings indicated above, bias-sewn tubing conforming to MIL-T-40625 may be used. One end of the tubing may be machine sewn. The formation of handles shall be effected by hand sewing in conformance with method as described above. Whenever machine sewing is used, the thread type, stitch type and number of stitches per inch shall conform to the requirements specified in MIL-T-40625.

5.2.1.2 Rigid containers. When specified by the procuring agency, tarpaulins of one type, class and size only, packaged as specified in 5.1 and packed in quantities as specified in table III, shall be packed in shipping containers that conform to overseas type, style A of PPP-B-601; style A, class 2, type 2 load of PPP-B-576; style 1, 2 or 2A, class 3 use, type 2 load of PPP-B-585; or class 2, style 2 or 4 of PPP-B-621. Each container shall be closed and reinforced in accordance with the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Cloth-covered bundles. Unless otherwise specified, tarpaulins shall be packed as specified in 5.2.1.1.

5.2.2.2 Rigid containers. When specified by the procuring agency, tarpaulins of one type, class and size only, packaged as specified in 5.1 and packed in quantities as specified in table III, shall be packed in shipping containers that conform to domestic type, style A or B of PPP-B-601; style A, class 1, type 2 load of PPP-B-576; style 1, 2 or 2A, class 1 use, type 2 load of PPP-B-585; or class 1, style 2 or 4 of PPP-B-621. Each container shall be closed and reinforced in accordance with the appendix of the applicable box specification.

5.2.3 Level C. Tarpaulins, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers, bundles and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2.4 All levels. As far as is practicable, packs shall be of such a size to lend themselves to palletization on 40 by 48 inch pallets. Finished packs will be permitted to overhang the pallet 1-1/2 inches on either side of the 40-inch width and 2 inches on either side of the 48-inch dimensions.

TABLE III. Paulin quantity per pack

Paulin area in square feet (sq. ft.)	Class 1 and 2 (each)	Class 3 (each)
0 to 45 sq. ft., inclusive	15	-
46 to 64 sq. ft., inclusive	10	-
65 to 120 sq. ft., inclusive	5	3
121 to 225 sq. ft., inclusive	3	3
226 to 400 sq. ft., inclusive	2	2
401 sq. ft. and over	1	-

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, cloth-covered bundles or rigid containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, cloth-covered bundles or rigid containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Uncoated tarpaulins. For purposes of this specification, noncoated tarpaulins are defined as those whose surfaces are not coated with natural or synethetic rubber, resins or other sealing compounds.

6.2 Intended use. Classes 1 and 2 tarpaulins are for heavy-duty cover, construction and general marine service use. Class 3 tarpaulins are for light-duty general tarpaulin use and for use where weight is a factor when transported by personnel.

6.3 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class and size required (see 1.2.1 and 1.2.2).
(c) Preproduction sample. When preproduction sample is not required (see 3.1).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3.1 For civil agency procurement, where level B packaging is specified in the contract or order, the requirements in 5.1.1 (level A) shall apply.

6.4 For access to a standard shade sample of thread, address the procuring office issuing the invitation for bids.

6.5 This specification covers uncoated, fire, water, weather and mildew-resistant tarpaulins (FWWMR) authorized for repetitive procurement in the Government services. All tarpaulins procured under this specification for use by military services will include webbing loops (rather than grommets) and will not include tie ropes.

6.6 A satisfactory method of applying the textile compound for the repair of leaking seams is as follows:

a. A suitable volume of textile compound conforming to type I of MIL-M-13295 (FSN 8030-264-3840) shall be thoroughly stirred and diluted with an equal volume of petroleum solvent (Stoddard solvent or equivalent). Thoroughly stir the mixture again before use.

b. Prior to application of the compound, the tarpaulins must be dry. Special attention must be taken to insure adequate ventilation and that proper fire preventive measures are taken since the compound is flammable.

c. Two coats of the textile compound shall then be applied to the seams (on the outside of the tarpaulin only). A four-inch wide stiff brush is recommended for use. The first coat shall be air dried for a minimum of four hours prior to application of the second coat. Elapsed time between coats may be reduced by providing a supply of hot air to evaporate the solvent more quickly. In this manner, a second coat may be applied as soon as the first coat is dry to the touch. After the application of the second coat, the treated tarpaulin shall be allowed to air dry for a period of at least 24 hours.

6.6.1 Seams to be treated. In the event of failure of the test specified in 4.4.1, treat all panel lap seams only.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 82

Review activities:

Army - MD
Navy - SA, YD
Air Force - 82, 85

User activities:

Navy - OS, SH, MC, CG

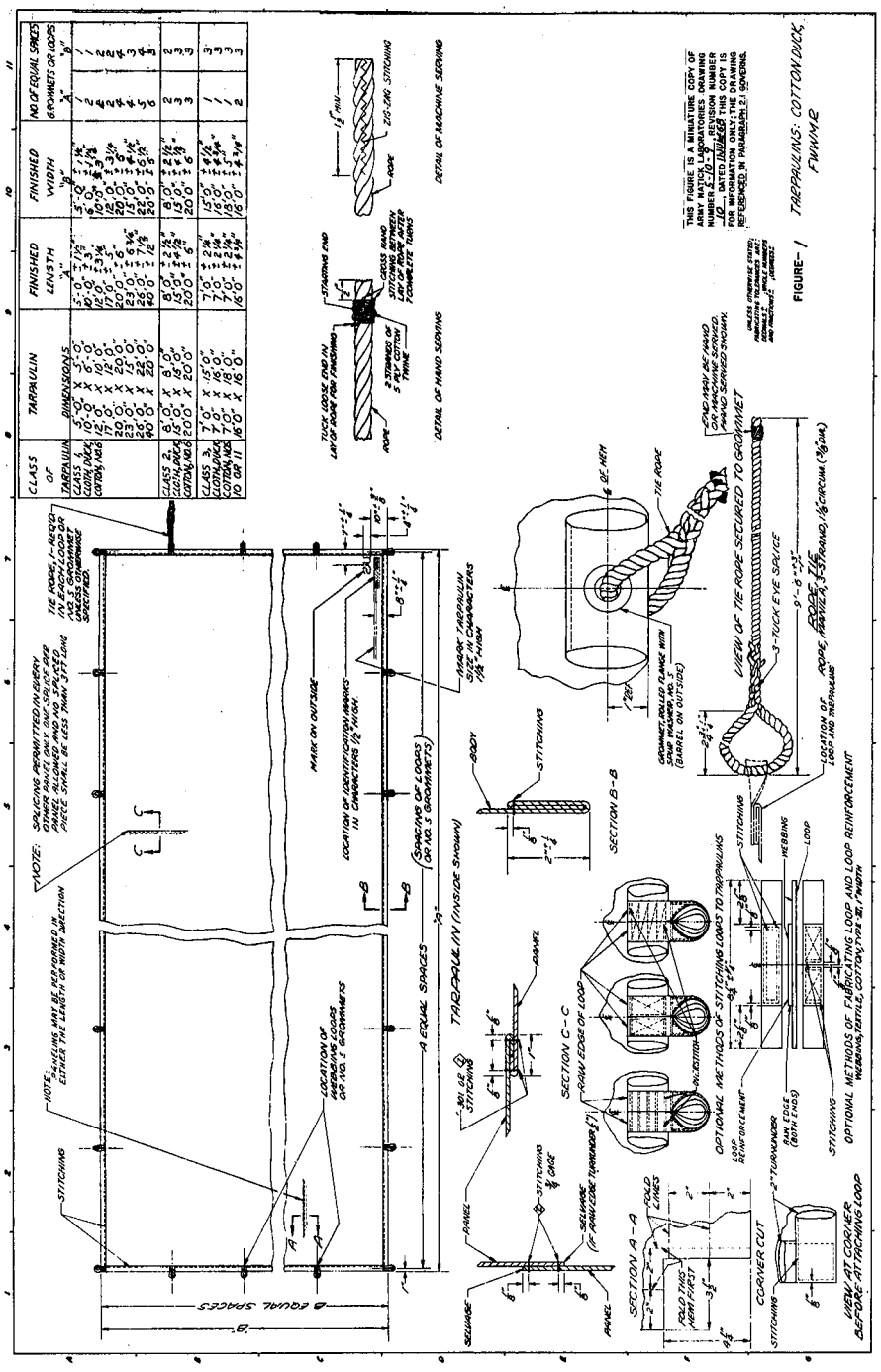
Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

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AGR
PO
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Tarpaulins: Cotton Duck; FWWMR K-P-146d

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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